EXT VANISHING AND HOM VANISHING ON REGULAR SCHEMES

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Local global dimension. algebra-proposition-regular-finite-gl-dim (cross-chapter, named). Stalkwise Ext vanishing above the local dimension.
- Cohomology vanishing. cohomology-proposition-vanishing-Noetherian (cross-chapter, named). Kills the spectral sequence terms.

Source reference. Original reference 00O7: focused-seeds/geometry/S9-eqv-Ext-0-regular/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0GM8: focused-seeds/geometry/S9-eqv-Ext-0-regular/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02UZ: focused-seeds/geometry/S9-eqv-Ext-0-regular/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BQP: focused-seeds/geometry/S9-eqv-Ext-0-regular/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0149: focused-seeds/geometry/S9-eqv-Ext-0-regular/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08J5: focused-seeds/geometry/S9-eqv-Ext-0-regular/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

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Source reference. Original reference 08E8: focused-seeds/geometry/S9-eqv-Ext-0-regular/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a regular Noetherian scheme of dimension $d < \infty$. Then*

(1) *for $\mathcal{F}$, $\mathcal{G}$ coherent $\mathcal{O}_X$-modules we have $\mathrm{Ext}^n_X(\mathcal{F}, \mathcal{G}) = 0$ for $n > d$, and*
(2) *for $K, L \in D^b_{Coh}(\mathcal{O}_X)$ and $a \in \mathbf{Z}$ if $H^i(K) = 0$ for $i < a + d$ and $H^i(L) = 0$ for $i \geq a$ then $\mathrm{Hom}_X(K, L) = 0$.*

Proof. To prove (1) we use the spectral sequence

$$H^p(X, \mathcal{E}xt^q(\mathcal{F}, \mathcal{G})) \Rightarrow \mathrm{Ext}^{p+q}_X(\mathcal{F}, \mathcal{G})$$

of Cohomology, Section 0BQP. Let $x \in X$. We have

$$\mathcal{E}xt^q(\mathcal{F}, \mathcal{G})_x = \mathcal{E}xt^q_{\mathcal{O}_{X,x}}(\mathcal{F}_x, \mathcal{G}_x)$$

see Cohomology, Lemma 0GM8 (this also uses that $\mathcal{F}$ is pseudo-coherent by Derived Categories of Schemes, Lemma 08E8). Set $d_x = \dim(\mathcal{O}_{X,x})$. Since $\mathcal{O}_{X,x}$ is regular the ring $\mathcal{O}_{X,x}$ has global dimension d_x, see Algebra, Proposition 00O7. Thus $\mathcal{E}xt^q_{\mathcal{O}_{X,x}}(\mathcal{F}_x, \mathcal{G}_x)$ is zero for $q > d_x$. It follows that the modules $\mathcal{E}xt^q(\mathcal{F}, \mathcal{G})$ have support of dimension at most $d - q$. Hence we have $H^p(X, \mathcal{E}xt^q(\mathcal{F}, \mathcal{G})) = 0$ for $p > d - q$ by Cohomology, Proposition 02UZ. This proves (1).

Proof of (2). We may use induction on the number of nonzero cohomology sheaves of K and L. The case where these numbers are $0, 1$ follows from (1). If the number of nonzero cohomology sheaves of K is > 1, then we let $i \in \mathbf{Z}$ be minimal such that $H^i(K)$ is nonzero. We obtain a distinguished triangle

$$H^i(K)[-i] \to K \to \tau_{\geq i+1} K$$

(Derived Categories, Remark 08J5) and we get the vanishing of $\mathrm{Hom}(K, L)$ from the vanishing of $\mathrm{Hom}(H^i(K)[-i], L)$ and $\mathrm{Hom}(\tau_{\geq i+1} K, L)$ by Derived Categories, Lemma 0149. Similarly if L has more than one nonzero cohomology sheaf. □